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Proposed System and Process Flow

1. Proposed system overview

We propose a centralized business management system for RCWheels Corporation covering requisition, inventory, purchasing, and delivery. Linked records will allow employees to follow an internal request from submission and approval through stock reservation, purchasing when needed, supplier receiving, internal delivery, and requester acceptance. The system will retain outstanding quantities and the history of each decision until the request is fulfilled or separately cancelled.

The proposal preserves separate approval responsibilities. The Authorized Manager, Department Head, or Boss approves internal requisitions according to assigned authority. Accounting Associates approve supplier purchases and changes to purchase commitments. Purchasing Staff evaluate suppliers and place approved orders by phone or message. The Inventory Custodian verifies stock, prepares issues, and inspects supplier receipts, with Operations or delivery personnel assisting in receiving and moving items. The requester inspects internally delivered items and signs the Acknowledgement Receipt after acceptance.

Scope of the proposed system

Prototype coverage is limited to the main location at New Imus, Lorega St., San Miguel, Cebu City. The other three branches, interbranch transfers and consolidated branch inventory are outside this prototype. Stock and reservations refer only to this location.

The Delivery module includes supplier receiving and internal delivery to employees within the main location, together with their related discrepancies, returns, replacements, and cancellations. Supplier receiving and internal delivery have separate flowcharts because they update different records. Customer dispatch, customer delivery scheduling, courier route planning, and interbranch transfers remain outside the proposed scope.

The four modules cover digital requests, stock balances and movements, reservations, item costs and values, Purchase Requisitions, Purchase Orders, supplier records, receiving records, internal dispatch, Acknowledgement Receipts, discrepancy reports, and in-system notifications. QuickBooks and paper or spreadsheet trackers are replaced only for the covered operational records. General ledger, tax, financial statements, sales processing, payments, and POS integration remain outside the project. No third-party application integration is included.

The proposed system also records manual stock deductions supported by sales records so shared inventory can be kept current without adding sales processing or payment handling. The Inventory Custodian records each deduction at stock release using an item, quantity, and unique source reference. Delayed entries or discrepancies require an incomplete-balance warning and physical verification before further allocation. These supporting stock controls remain separate from the illustrated requisition-to-delivery sequence.

2. Proposed requisition process

The requisition module records internal requests for supplies and materials. Authorized employees complete the digital Requisition Form (RF); authorized personnel submit on behalf of mechanics and identify both the submitter and the employee who needs the items. The system records the need, validates the entry, checks stock information, and routes the request to the assigned approver. An urgent request receives priority attention but still requires approval.

Process sequence

1. An authorized employee enters the item, quantity, purpose, date needed, and requester. A request marked urgent includes the reason for urgency.

2. The system validates required fields and the submitter role. Invalid entries return to the employee with errors highlighted. After a valid submission is saved, the system returns a unique requisition ID and timestamp.

3. The system checks recorded usable on-hand stock and active reservations. If the recorded quantity is insufficient, it notifies the requester and relevant staff and retains the request for manager review. Both sufficient-stock and shortage requests proceed to approval. Urgent requests are prioritized before the same manager-review step.

4. The assigned manager reviews the need and stock information. A rejection records the reason and notifies the requester in-system. The requester may correct and resubmit the request through field validation and approval again, or leave it rejected. Earlier decisions remain in the history.

5. The system saves the approval and passes the approved requisition to Inventory through connector A. An approved request waiting for stock uses Awaiting Stock status. Approval of the requisition does not approve a supplier purchase; a stock shortage must proceed through a linked Purchase Requisition and separate purchase approval.

6. The system tracks quantities requested, reserved, dispatched, accepted, returned, and still outstanding. A dispatch leaves the request open until the requester accepts the items. Partial acceptance or a replacement need keeps the remaining requirement open. The manager marks the requisition Fulfilled only after all required quantities are accepted, unless an outstanding balance is separately cancelled with authorization.

Records and expected improvement

The module retains request details, urgency, approval or rejection, actor identities, timestamps, quantity progress, and links to purchases, dispatches, receipts, and discrepancies. Mandatory fields and retained status history are expected to reduce incomplete submissions and repeated requests for updates; no measured time saving is claimed.

Cancellation is separate from fulfillment and rejection. The requisition approver may cancel an outstanding balance with a recorded reason. The system releases only unused reservations, retains accepted and dispatched history, and reviews related purchase commitments separately. A requisition cancellation does not automatically cancel a supplier PO. Items rejected during internal delivery enter Pending Resolution; they do not turn the original approval into a rejected requisition.

3. Proposed inventory process

Inventory maintains a common record of usable stock, reservations, accepted receipts, requisition issues, returns, and approved adjustments. The Inventory Custodian performs physical verification and prepares stock issues. The Inventory Control or Monitoring Manager oversees stock records and authorized corrections. The module receives approved requests through A, notifications of posted supplier receipts through D, and remaining or replacement needs through F.

1. Load open approved requests and calculate the unissued requirement after accounting for accepted and in-transit quantities. A supplier receipt arriving through connector D has already been posted by Supplier Receiving. Inventory must not post the same receipt again. If no further issue is needed now, finish the inventory check while retaining any pending delivery and its acceptance requirement.

2. Check recorded stock and physically verify usable quantities. For a new request, available stock is usable on-hand quantity less active reservations. For an existing request, include its own valid reservation and exclude reservations belonging to other requests. Recount mismatches, review stock movements and reservations, and obtain the required authorization before correcting stock records. Recheck availability after the correction.

3. Reserve only the additional quantity needed, retaining valid existing reservations and protecting quantities reserved for other requests. The Custodian may prepare a full or partial issue. The Custodian prepares the digital Acknowledgement Receipt (AR), corrects invalid fields, and submits the validated AR for storage. At this stage the AR is prepared for delivery; the requester signs it only after accepting the items.

4. When items are ready, the manager sets For Delivery. At actual dispatch, the system records the issue once, reduces usable warehouse on-hand stock by the dispatched quantity, consumes the matching reservation, and records the quantity in transit. It notifies the requester that the items are out for delivery. Connector C sends the issued items and AR to Internal Delivery. Acceptance later updates request progress without a second stock deduction.

5. Independently of the issued quantity, calculate any uncovered shortfall. Exclude quantities already accepted, dispatched, reserved, or covered by allocated outstanding purchase commitments. If the shortfall is positive, the Custodian creates a linked digital Purchase Requisition (PR) and sends it to Purchasing through connector B. Otherwise, retain the open request and await committed stock or delivery acceptance. Use Awaiting Stock for approved quantities waiting for replenishment.

6. Check stock after a stock or reservation change and during routine reviews. Available stock is usable on-hand stock minus active reservations. If available stock is less than or equal to the item reorder point, notify the Custodian and Inventory Manager. The Custodian verifies stock and checks pending PRs and open POs before calculating the extra quantity needed. If this quantity is greater than zero, create a PR and send it through B for purchase approval. Otherwise, follow up existing purchases and continue monitoring. The Inventory Manager maintains each item reorder point. The reorder point tells staff when to review stock; it is not the order quantity.

Safeguard for stock shared with sales

Unrecorded sales can make the ledger overstate available stock. The Custodian records sales-related deductions at release using supporting sales records and unique source-document lines. A source line can be posted only once; corrections use a linked reversal and replacement entry. Stock reserved for another approved request cannot be consumed by an unrelated sale. Delayed entries or discrepancies mark the affected balance incomplete and require physical verification before allocation. Count-time adjustments alone do not keep balances current between counts.

The Inventory Control or Monitoring Manager reviews supporting references and exceptions. Weekly or monthly physical counts remain a separate scheduled activity: recount mismatches, review evidence, and approve corrections with a reason and an audit record. A variance must not be classified as sales without support. Damaged or uncertain stock remains unavailable for normal issue. Sales processing and POS integration remain outside the system.

4. Proposed purchasing process

Purchasing receives PRs for an uncovered request shortfall or verified stock replenishment from Inventory through B. It also receives G from Supplier Receiving when cancelled or failed supply requires a review of the remaining demand. Purchasing Staff select suppliers and place orders, while Accounting Associates separately approve purchase commitments. Any separately reviewed replenishment need must follow the same purchase-approval controls.

1. Load the PR and verify the uncovered quantity against current stock, reservations, in-transit issues, and allocated purchase commitments. For a G handoff, review the remaining need with the requisition approver. If the remaining request is separately cancelled, record the authorization, release unused reservations, and notify the requester. If the need remains, review the linked PR and source the uncovered quantity without duplicating existing commitments.

2. Purchasing Staff evaluate suppliers against price, quality, availability, and reliability. An unsuitable supplier returns the process to supplier evaluation. Staff prepare a Purchase Order (PO) linked to the PR, including items, quantities, prices, supplier, and any related requisition reference.

3. Save the proposed PO and submit it to Accounting Associates for review. If the budget is insufficient, Accounting records the reason and marks the PO For Revision. The system notifies Purchasing Staff, who revise and resubmit it. Every revised PO returns to Accounting for review before commitment; a revision status alone does not authorize an order.

4. When Accounting approves the PO, save the approver, timestamp, and approved version. Purchasing Staff place the approved order by phone or message and record the supplier commitment. The system does not independently select suppliers, authorize spending, or send an unapproved order.

5. Retain the approved quantities, prices, supplier, order date, expected delivery date, and links to the PR and requisition where applicable. An approved change to supplier, item, quantity, or price requires Accounting reapproval before the revised commitment is placed. Purchase and requisition approvals remain separate.

6. Connector E sends the approved PO and expected quantities to Supplier Receiving. Purchasing Staff follow up on expected delivery dates and retain revised commitments without losing the earlier history. A cancelled PO balance does not erase accepted receipts and does not automatically cancel the related requisition.

Controls and expected improvement

Supplier selection and purchase approval remain human responsibilities. Purchasing receives PRs for uncovered approved requests and for verified replenishment when available stock reaches or falls below the reorder point. Pending PRs and outstanding PO quantities must be checked before creating additional demand. A low-stock alert or PR does not authorize expenditure; Accounting Associates must approve the PO. Generated draft POs remain optional future work.

Linked PRs, POs, supplier information, revision history, and receiving balances are expected to reduce repeated transcription and make delivery checks easier. They do not guarantee supplier availability, stable prices, or on-time delivery. If a purchase is cancelled rather than revised, retain its reason and keep the underlying demand open until it is covered or separately cancelled.

5. Proposed delivery process

The Delivery module covers two related processes within the main location: receiving supplier deliveries and delivering issued items to internal requesters. Supplier Receiving uses the approved PO, supplier Delivery Receipt (DR), and invoice. Internal Delivery uses the prepared Acknowledgement Receipt (AR) and the dispatched requisition quantities. Customer delivery scheduling remains outside the scope.

5.1. Supplier receiving

1. Supplier Receiving opens the approved PO received through connector E and records the arriving supplier delivery. The Custodian checks the DR, invoice, actual quantities, specifications, and condition against the outstanding PO lines. Operations or delivery personnel may assist with receipt and movement of the goods.

2. Identify accepted, rejected, missing, excess, and disputed quantities separately. Partial acceptance is permitted for conforming quantities. Lateness or a short delivery does not by itself prevent accepting correct goods. Hold rejected or uncertain items outside usable stock while their resolution is pending.

3. If any quantity is accepted, the Custodian signs the DR for that quantity. Save the receiving record and its source reference once, increase usable stock only by the accepted quantity, and update the received and outstanding quantities per PO line. If nothing is accepted, record the rejection without posting usable incoming stock.

4. Connector D notifies Inventory that accepted stock is already posted and can be considered for waiting approved requests. Inventory preserves existing reservations and rechecks the unissued requirement before further allocation. A receipt alone does not fulfill the employee requisition, and the D handoff must not produce another stock inflow.

5. Reconcile the outstanding PO balance and discrepancies. If a balance or discrepancy remains, the Custodian issues a Supplier Discrepancy Report (SDR) identifying the reason, evidence, affected quantities, and requested action. Purchasing Staff coordinate with the supplier and record the expected replacement or follow-up delivery date. Each follow-up delivery repeats inspection against the remaining PO balance.

6. Close receiving only when all ordered quantities have been accepted or the affected balance has been formally cancelled and discrepancies are resolved. Accounting Associates authorize cancellation of outstanding purchase commitments with a recorded reason. Confirm cancellation with the supplier, arrange return of rejected goods held, update the PO balance, and use G to review any remaining requisition need with Purchasing. Cancellation of supplier supply does not itself cancel the requisition.

Exception treatment

Each accepted receipt has a unique source reference so a follow-up shipment cannot repost an earlier accepted quantity. Supplier discrepancies remain open until a resolution is agreed and recorded. Excess quantities require an explicit decision rather than automatic addition to usable stock. A replacement becomes incoming stock only after a new inspection and accepted receipt.

For a supplier return, retain the original receipt reference, quantity, reason, and supporting documents. Reverse stock and its recorded inventory value only if that quantity was previously posted. Goods rejected before stock posting cause no stock-value reversal. Purchasing Staff retain the supplier commitment or credit or refund reference; processing payments, credits, refunds, and statutory accounting entries remains outside the system.

A fully resolved PO may close with zero accepted quantity if all outstanding quantities are formally cancelled. The linked requisition remains open for replacement or other supply unless it is fulfilled or separately cancelled by its approver. Expected delivery dates and unresolved quantities support follow-up; physical inspection, supplier contact, and receipt signatures remain human responsibilities.

5.2. Internal delivery and returns

Internal Delivery receives the dispatched items and prepared AR from Inventory through connector C. The requester checks the delivered items before acceptance. The flow retains the existing requisition approval and resolves delivery problems through Pending Resolution, replacement, or cancellation. It does not treat every returned item as damaged.

1. Delivery personnel present the items for inspection. The requester checks quantity, specification, and condition against the approved request and dispatched items. Final handover follows acceptance.

2. If the items are acceptable, the requester accepts them and signs the digital AR. The system saves the signed AR and accepted quantities and clears the corresponding in-transit quantity. The requester notifies the manager, who confirms progress. No second stock deduction is made at acceptance.

3. If an approved quantity remains unfulfilled, connector F returns the remaining need to Inventory. Inventory subtracts quantities already accepted or still in transit before preparing another issue. When all required quantities are accepted, the manager marks the request Fulfilled, and the system saves the completed requisition.

4. If items are unacceptable, the requester issues a Requester Discrepancy Report (RDR) with the requisition ID, reason, evidence, affected quantities, and requested action. The system sets Pending Resolution, notifies the manager, and arranges the return. Delivery personnel return the affected items to the designated stockroom and storage area.

5. Record each return once and clear its returned in-transit quantity. Hold items unavailable until inspected. Restore a confirmed usable return to usable warehouse stock; keep damaged or defective stock separately unavailable. Uncertain items remain held for reinspection. The return must not be recorded again each time its condition is reviewed.

6. The manager reviews the discrepancy and authorizes replacement or cancellation. Replacement keeps the same requisition open and uses F to resume Inventory checking; an unavailable replacement proceeds through the normal shortage and purchasing controls. Cancellation records the reason, releases unused reservations, notifies the requester, and closes only the affected remaining request.

Forms and records

The process uses the RF, PR, PO, AR, supplier DR, supplier invoice, SDR, and RDR. RF, PR, PO, AR, SDR, and RDR are prepared internally; the DR and invoice are supplied by the supplier. Use DR only for Delivery Receipt and use SDR and RDR to distinguish the discrepancy reports. Stock movements, reservations, dispatches, and returns are linked records rather than additional paper forms.

6. Implementation decisions

Opening balances and initial operation

Start the system with verified current stock quantities and values and the supplier list. Historical transactions are not migrated. The Custodian prepares opening quantities and proposes stock adjustments with evidence. The Inventory Control or Monitoring Manager verifies quantities, maintains item records, and approves quantity corrections. Accounting Associates verify opening values and approve changes to recorded inventory value. Purchasing Staff maintain supplier records. Requisition approvers, purchase approvers, custodians, receiving personnel, and requesters use individual accounts with an audit trail.

Implement central records, digital RFs and PRs, separate approvals, reservations, supported sales-stock deductions, POs, supplier receipts, ARs, internal dispatch and acceptance, discrepancies, returns, traceability, and in-system notifications as the core workflow. Verify each receipt, reservation, dispatch, return, and acceptance before extending automation. Include item reorder points and in-system low-stock alerts with physical verification and checks against pending PRs and outstanding POs. Generated draft POs remain optional later work.

Proposed operating controls

Branch scope: implement the four modules for the main location only. A future extension to other branches requires separately identified stock locations, branch-specific balances, reservations and transfer controls.

1. Sales deductions and shared stock: record supported deductions at stock release using unique source-document lines. Delayed entries trigger incomplete-balance warnings and physical verification. The Inventory Control or Monitoring Manager reviews exceptions. Sales processing and POS integration remain excluded.

2. Stock controls: reserve only verified usable quantities for approved requests; permit partial issue and partial acceptance; preserve other reservations; and prevent duplicate replenishment. Post supplier receipts once in Supplier Receiving. Deduct issued stock once at dispatch and clear only the matching reservation. Keep accepted, in-transit, returned, and outstanding quantities distinct.

3. Approval and exceptions: retain separate requisition and purchase approval. PO changes to supplier, item, quantity, or price require reapproval. Accounting authorizes purchase-balance cancellation; the requisition approver separately authorizes cancellation of remaining request quantities. Partial receipts and partial internal acceptance keep outstanding quantities open. Classify usable, damaged, and uncertain returns before updating availability.

4. Records and access: verify opening quantities and values, retain supporting references, and use individual accounts with role-based access. The academic prototype requires browser access on a computer. Database access is shared through the central store but limited by each user role; DB1 to DB5 represent module connections rather than five separate databases.

5. Notifications: use in-system updates for submission errors, urgency, approval, rejection, stock shortages, PO revision, purchasing, accepted supplier receipts, actual dispatch, internal discrepancies, cancellation, and fulfillment. Automated email notifications remain outside the project scope.

6.1. Central database and module connections

The proposed system uses one central database for all four modules and the two Delivery workflows. DB1 to DB5 are module-specific connection references: DB1 for Requisition, DB2 for Inventory, DB3 for Purchasing, DB4 for Supplier Receiving, and DB5 for Internal Delivery and Returns. Each matching reference accesses the shared store subject to role permissions; it is not a separate database.

The proposed logical tables cover users and roles; requisitions, lines, and status history; items including reorder points, balances, reservations, movements, and adjustments; suppliers, PRs, POs, and their lines; supplier receipts, discrepancies, and commitments; dispatches, ARs, internal discrepancies, returns, and return lines; and notifications, document attachments, and the audit log. The database diagram identifies these proposed table names. Detailed keys, relationships, and physical implementation are finalized during database design.

Store item-level quantities and source references so receipts, issues, returns, and sales-supported deductions cannot be posted twice. Link transactions to their requisition, PR, PO, receipt, dispatch, or return as appropriate. Retain the actor, timestamp, reason, and authorized status changes. Reservation and dispatch updates must be consistent so simultaneous requests cannot claim the same available stock.

Process connector directory

A: Requisition to Inventory - approved requisition.

B: Inventory to Purchasing - PR for an uncovered request shortfall or verified reorder-point replenishment.

C: Inventory to Internal Delivery - dispatched items and prepared AR.

D: Supplier Receiving to Inventory - accepted stock already posted.

E: Purchasing to Supplier Receiving - approved PO and ordered quantities.

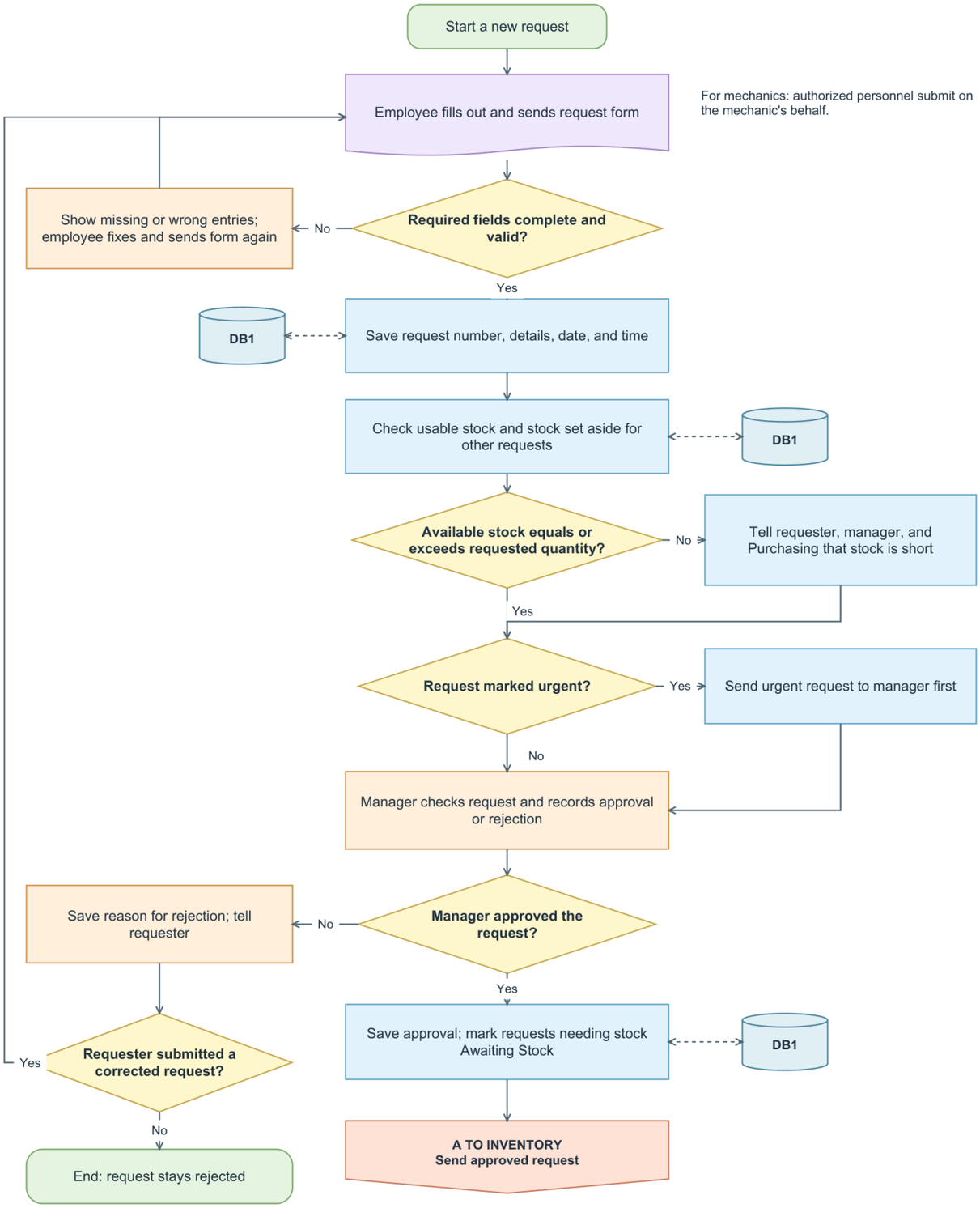
F: Internal Delivery to Inventory - remaining or replacement need.

G: Supplier Receiving to Purchasing - cancelled or failed supply requiring review of remaining demand.

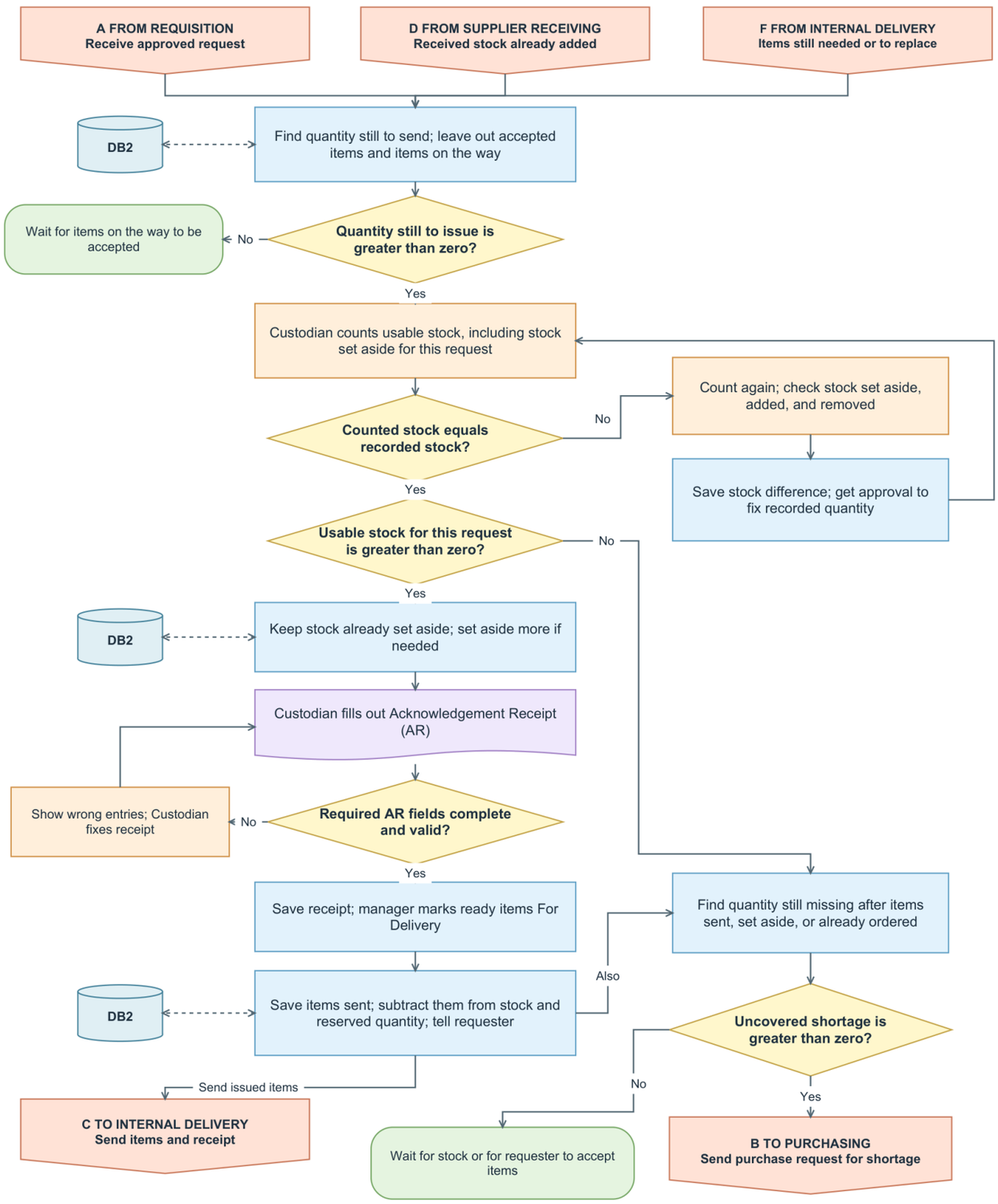
In the figures, purple shapes identify documents, blue rectangles identify system processing, orange rectangles identify human actions, yellow diamonds identify decisions, and green rounded shapes identify a start, end, or waiting state. Peach pentagons A to G identify process handoffs. Blue-gray cylinders DB1 to DB5 identify database connections; dotted arrows represent data exchange. Some action boxes summarize an actor action and its recorded system update.

7. Flowcharts

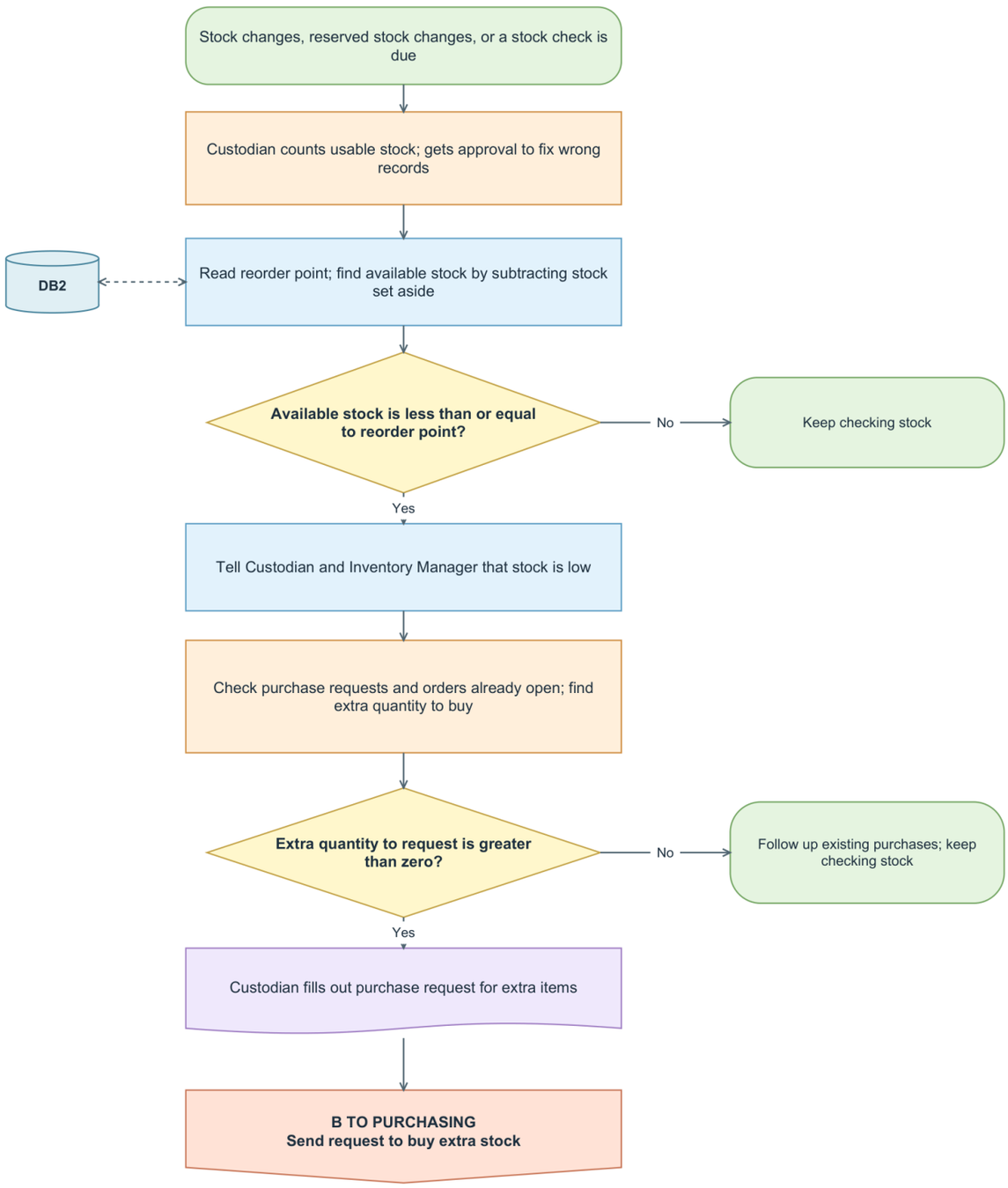
7.1. Requisition



7.2. Inventory



7.3. Inventory Reorder Point



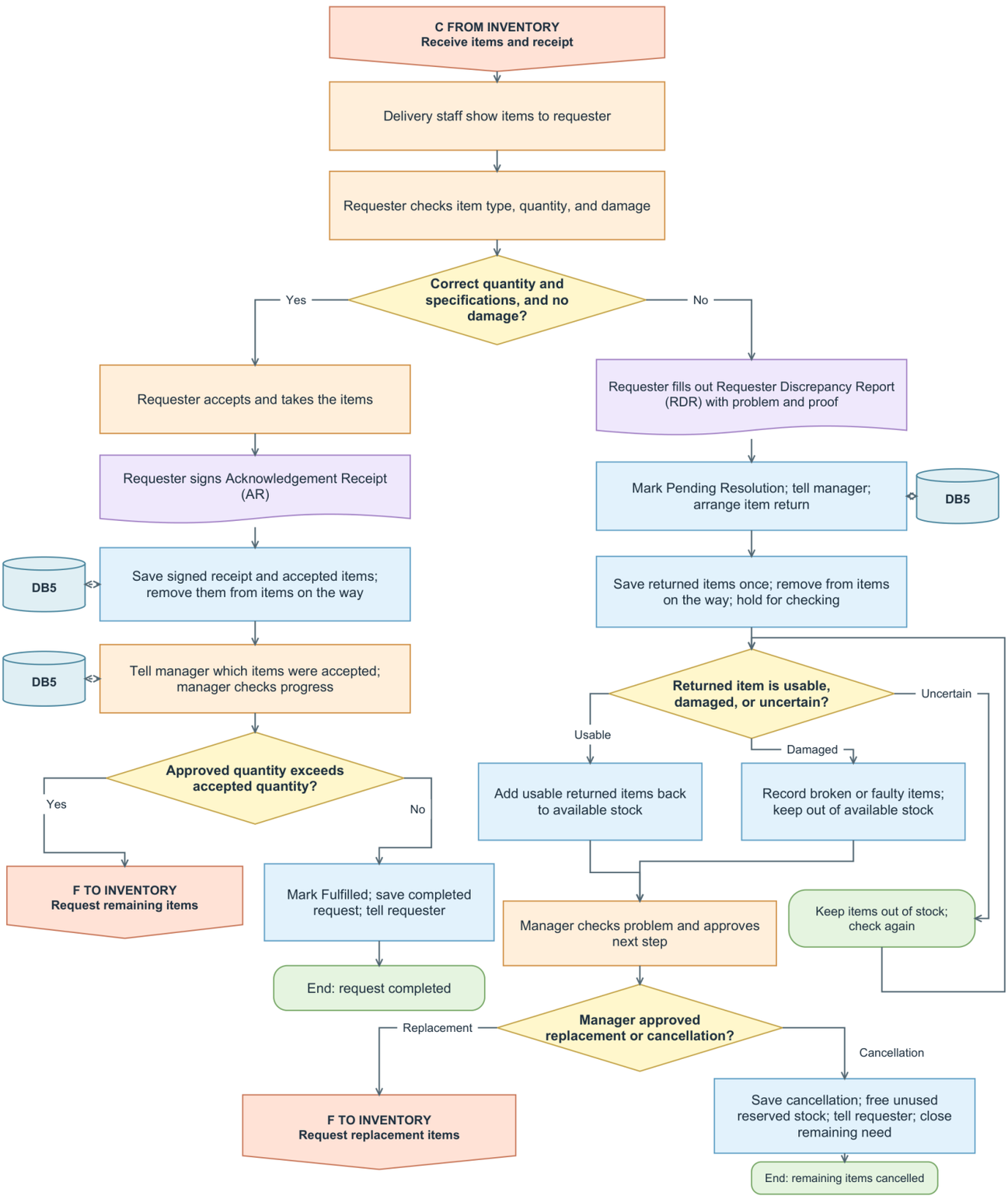
7.4. Purchasing



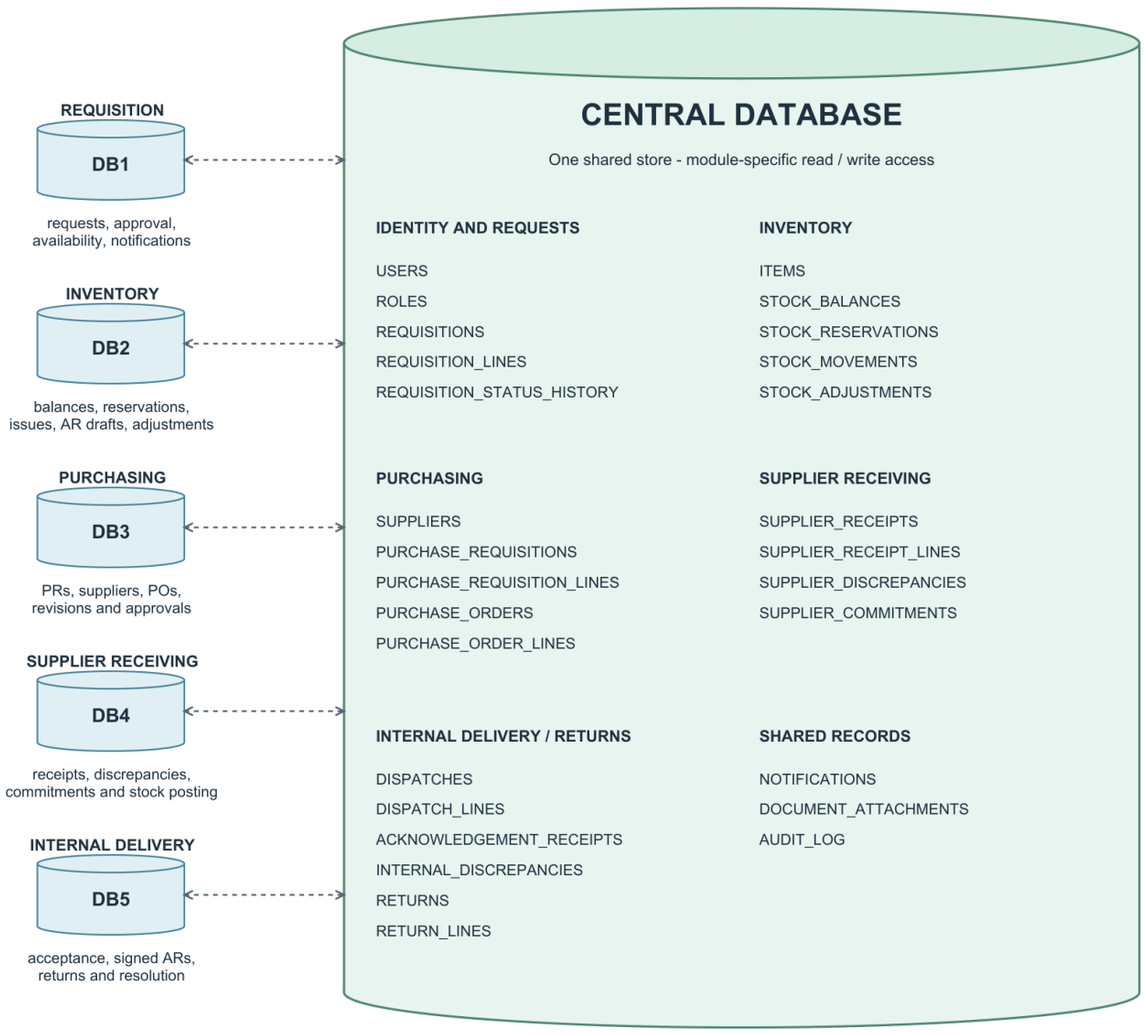
7.5. Supplier Receiving



7.6. Internal Delivery and Returns



7.7. Central Database



PROCESS CONNECTOR DIRECTORY

A	Requisition -> Inventory	Approved requisition
B	Inventory -> Purchasing	Shortfall / replenishment PR
C	Inventory -> Internal delivery	Dispatched items and AR
D	Supplier receiving -> Inventory	Accepted stock already posted
E	Purchasing -> Supplier receiving	Approved PO / supplier order
F	Internal delivery -> Inventory	Remaining or replacement need
G	Supplier receiving -> Purchasing	Cancelled supply; review remaining demand